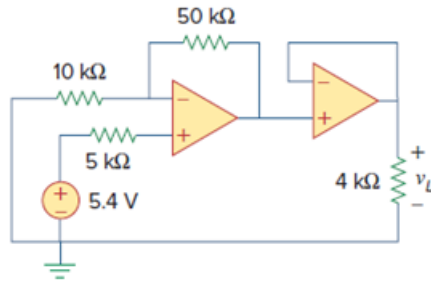


Problem

Determine the load voltage v_L in the circuit of Fig. 5.99.

Figure. 5.99:



Step-by-step solution

Step 1 of 3

It is given that a circuit as shown in figure 1.

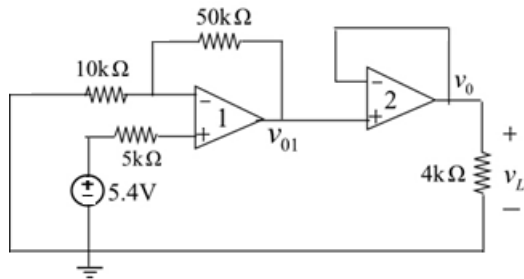


Figure 1

Step 2 of 3

Consider that the op-amp 1 in figure 1 is a non-inverting amplifier.

Therefore, the output of the op-amp 1 is

$$v_{o1} = \left(1 + \frac{R_f}{R} \right) v_i$$

Here, $R_f = 50\text{ k}\Omega$, $R = 10\text{ k}\Omega$ and $v_i = 5.4\text{ V}$.

Substitute the values.

$$\begin{aligned} v_{o1} &= \left(1 + \frac{50\text{ k}\Omega}{10\text{ k}\Omega} \right) 5.4 \\ &= 32.4\text{ V} \end{aligned}$$

Step 3 of 3

Consider that the op-amp 2 in figure 1 is a voltage follower.

Therefore, the output of the op-amp 2 is,

$$v_o = v_{o1}$$

Substitute 32.4 V for v_{o1} .

$$v_o = 32.4\text{ V}$$

Therefore, the load voltage v_L is $\boxed{32.4\text{ V}}$.